

Diag

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 40 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 13$ cm. The length of PQ is:
(A) (A) 10 cm
(B) (B) 8 cm
(C) (C) 12 cm
(D) (D) 7 cm
2. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° , then $\angle POA$ is equal to:
(A) (A) 50°
(B) (B) 60°
(C) (C) 70°
(D) (D) 80°
3. From a point Q , the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is:
(A) (A) 9 cm
(B) (B) 12 cm
(C) (C) 15 cm
(D) (D) 7 cm
4. Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is:
(A) (A) 10 cm
(B) (B) 8 cm
(C) (C) 6 cm
(D) (D) 4 cm

5. The number of tangents that can be drawn from a point inside a circle is:
- (A) (A) 1
 - (B) (B) 2
 - (C) (C) 0
 - (D) (D) Infinite
6. If the circumference of a circle is 176 cm, then its radius is:
- (A) (A) 14 cm
 - (B) (B) 21 cm
 - (C) (C) 28 cm
 - (D) (D) 35 cm
7. The area of a circle whose diameter is 14 cm is:
- (A) (A) 44 cm^2
 - (B) (B) 154 cm^2
 - (C) (C) 616 cm^2
 - (D) (D) 308 cm^2
8. The area of a quadrant of a circle with radius 7 cm is:
- (A) (A) 38.5 cm^2
 - (B) (B) 77 cm^2
 - (C) (C) 154 cm^2
 - (D) (D) 19.25 cm^2
9. If the perimeter of a semi-circular protractor is 36 cm, then its diameter is:
- (A) (A) 7 cm
 - (B) (B) 21 cm
 - (C) (C) 28 cm
 - (D) (D) 14 cm
10. The area of a sector of a circle with radius 6 cm and central angle 60° is:
- (A) (A) $3\pi \text{ cm}^2$
 - (B) (B) $4\pi \text{ cm}^2$
 - (C) (C) $6\pi \text{ cm}^2$
 - (D) (D) $12\pi \text{ cm}^2$
11. Assertion (A): The length of a tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.

- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
- 12.** Assertion (A): If the distance between the centres of two circles of radii r_1 and r_2 is d , then they touch each other externally if $d = r_1 + r_2$. Reason (R): The distance between the centres of two circles is the sum of their radii when they touch internally.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
- 13.** Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A tangent to a circle intersects the circle at only one point.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
- 14.** Assertion (A): The length of the tangent from a point P at a distance of 10 cm from the centre of a circle of radius 6 cm is 8 cm. Reason (R): In a right triangle, the square of the hypotenuse equals the sum of the squares of the other two sides.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
- 15.** Assertion (A): Two parallel tangents can be drawn to a circle at the ends of a diameter. Reason (R): Tangents at the ends of a diameter are perpendicular to the diameter.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
16. Assertion (A): The area of a circle with radius 7 cm is 154 cm^2 . Reason (R): The area of a circle is given by πr^2 , where r is the radius.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
17. Assertion (A): The circumference of a circle with diameter 14 cm is 44 cm. Reason (R): The circumference of a circle is given by πd , where d is the diameter.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
18. Assertion (A): The area of a semicircle with radius r is $\frac{1}{2}\pi r^2$. Reason (R): The circumference of a semicircle is πr .
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
19. Assertion (A): If the perimeter and the area of a circle are numerically equal, then the radius of the circle is 2 units. Reason (R): Circumference $2\pi r = \pi r^2$ implies $r = 2$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

20. Assertion (A): The area of a sector of a circle with radius r and central angle θ is $\frac{\theta}{360} \times \pi r^2$. Reason (R): The length of the arc of a sector is $\frac{\theta}{360} \times \pi r^2$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

- Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
- If a tangent PA drawn from an external point P to a circle with center O makes an angle of 30° with the chord AB , where AB is the diameter, find $\angle APB$.
- In the given figure, if $\angle OAB = 40^\circ$, find $\angle ACB$ where O is the center of the circle.
- A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm and $CD = 4$ cm, find the length of AD .
- Two tangents TP and TQ are drawn to a circle with center O from an external point T . If $\angle PTQ = 70^\circ$, find $\angle POQ$.
- Find the area of a sector of a circle with radius 6 cm if the angle of the sector is 60° . (Use $\pi = 3.14$)

Section C: Short Answer (3 marks each)

- The angle of elevation of the top of a chimney from the top of a tower is 30° and the angle of depression of the foot of the chimney from the top of the tower is 60° . If the tower is 40 m high, find the height of the chimney.
- A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.
- Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
- If a tangent PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° , then find $\angle POA$.
- Prove that the lengths of tangents drawn from an external point to a circle are equal.
- A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 12$ cm. Find the length PQ .

Section D: Long Answer (4-5 marks each)

1. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30° , which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60° . Find the time taken by the car to reach the foot of the tower from this point.
2. The angle of elevation of the top of a tower from a point on the ground is 30° . On walking 40 metres towards the tower, the angle of elevation changes to 60° . Find the height of the tower and the distance of the initial point from the tower.
3. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height 6 metres. At a point on the plane, the angle of elevation of the bottom and top of the flagstaff are 30° and 60° respectively. Find the height of the tower.
4. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower.
5. Prove that the lengths of tangents drawn from an external point to a circle are equal.
6. PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q intersect at a point T . Find the length of TP .
7. A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.
8. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.